

Course: B. Tech.

Semester : V

Branch : Electrical and Power/Electrical & Electronics Engineering/Electrical Engineering

Subject Code & Name: Power System Analysis (BTEEC501)

Max Marks: 60

Date: 01/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

12

A) How will you draw a reactance diagram when the single line diagram of a power system is given?

Level 1

6

B) Draw reactance diagram for power system use a base of 50 MVA, 20 kV. On generated side the rating of generator, motor and transformer are:

Level 2

6

Generator = 20 MVA, 20 kV, $X = 20\%$

Motor = 35 MVA, 13.2kV, $X = 25\%$

Transformer 1, T1 = 25 MVA, 18/230 kV (Y/Y), $X = 10\%$

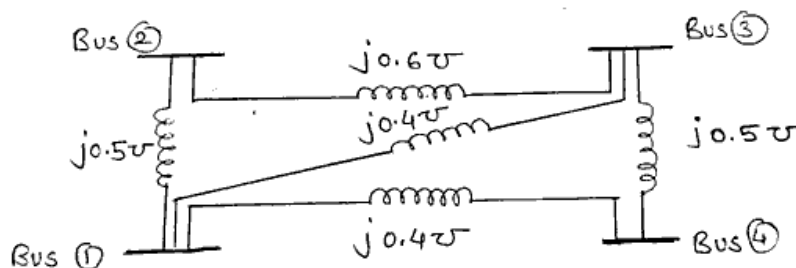
Transformer 2, T2 = 45 MVA, 230/13.8 kV (Y/ Δ), $X = 15\%$

Transmission Line = $j 30 \Omega$

C) Determine the Y_{BUS} matrix using Direct method.

Level 1

6



Q.2 Solve Any Two of the following.

12

A) Determine Y_{bus} by using Singular Transformation Method.

Level 1

6

Element No.	Bus Code PQ	Self-Admittance (p.u.)
1	0-1	1.2
2	1-2	1.6
3	2-3	2.0
4	2-0	1.4
5	3-0	2.2

B) Classify the various types of buses in a power system for load flow studies.

Level 1

6

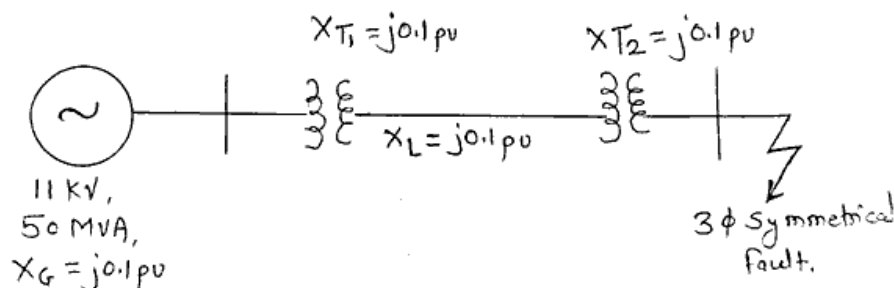
State the significance of reference buses.

C) Draw and Explain Automatic generation Control System.	Level 2	6
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Q.3 Solve Any Two of the following.		12
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A) Describe the construction of bus impedance matrix (Z_{BUS}) using building algorithm for lines without mutual coupling.	Level 2	6
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B) Find the fault current and short circuit MVA of the given power system. Consider 50 MVA as Base MVA & a 3 ϕ symmetrical fault occur at load side.	Level 1	6
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C) Explain sudden short circuit of 3 ϕ unloaded alternator by using approximate alternator model.	Level 2	6
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Q.4 Solve Any Two of the following.		12
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A) What are the effects of faults in power system? Explain symmetrical fault and unsymmetrical fault.	Level 1	6
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B) Two generator A & B are identical and rated at 11 kV & 10 MVA and have transient reactance of 20% at their own MVA base. Transformer 1 and Transformer 2 are also identical and are rated 5 MVA, 11/66 kV and have reactance of 5% at their own MVA Base. The tie line is 100 km long and has a reactance of 0.1 Ω /km/ phase. A 3 ϕ fault occurs at a distance of 25 km from one end of line when the system is on no load but rated voltage. Calculate fault current and fault MVA.	Level 1	6
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C) Derive the expression for fault current and draw the interconnection of sequence networks for double line to ground fault on the terminals of an unloaded generator.	Level 2	6
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Q.5 Solve Any Two of the following.		12
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A) Write a note on Contingency Analysis.	Level 1	6
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B) A single phase load of 100 KVA is connected across lines B & C of 3 ϕ supply of 3.3 kV. Determine symmetrical components of line currents.	Level 2	6
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C) What is Power Quality of power system? What are the causes of power quality issues?	Level 1	6
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